

DATAFORGE 2026 • PATHWAY • CLOUD 9 • KEY-VALUE CACHING VS RECURRENT MEMORY

# Why forgetting is the *price of remembering cheaply*.

Honest version: Transformers are that friend who saves every screenshot. Great memory, dead phone storage. Recurrent models are the friend who keeps one whiteboard and wipes it. We built this at KGP because we kept arguing about which friend BDH is. Spoiler: it picked the whiteboard — on purpose. Come break it with us.

OUR BET • PROVE US WRONG

**KV-cache: one drawer per word.** Grows forever, never lies. **Whiteboard: N slots, total.** Cheap forever, quietly lies when two words want the same corner. If you can keep the whiteboard honest at 500 words in 16 slots, we lose.

Built by Cloud 9, DataForge. If you know attention a little, you're good. 6 mins. The toy below is ours — scrappy, live, definitely not the real BDH.

HOW THIS PAGE WORKS

## No lecture. Just mess with it.

### 1 • STARE

It's already running — 120 pairs, 32 slots. Cabinet's chilling at 100%. Whiteboard's already messed up ~80 times. Click a yellow cell. Ask it what happened.

### 2 • BE MEAN

Press "Flood it". That's 500 words into 16 slots — hostel-wifi-energy. Watch the whiteboard give up. Then be nice: "Give it room".

3 • OH, SO BDH DID THIS TOO

Yeah. The real machine chose the whiteboard life. Same headache, fancier math. Scroll and we'll show you the receipts.

THE LAB • LIVE TOY

## Two memories, one stream of garbage words.

We feed both the same junk — mango-3 → 42, chai-7 → 88, you get it. Cabinet files everything. Whiteboard goes `hash(key) % N` and whoever wrote last wins. Old IIT trick: last entry in the mess register counts. Pick a key below and catch the whiteboard lying.

PAIRS STREAMING IN • N



500 pairs

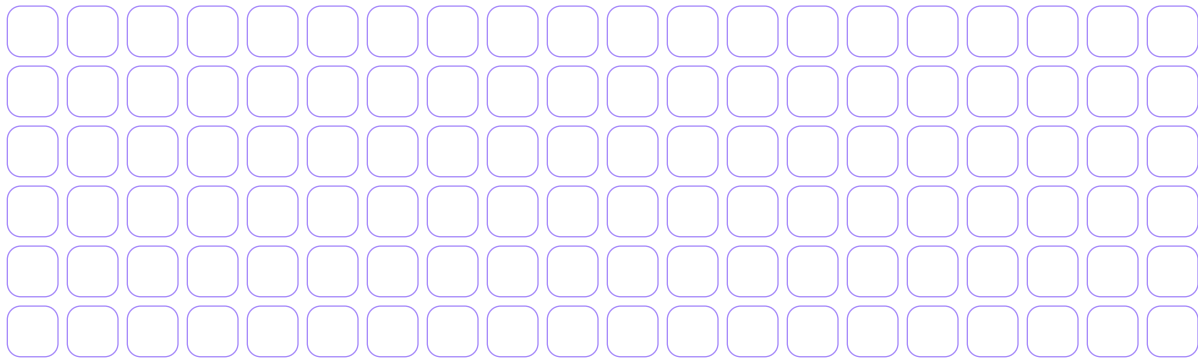
WHITEBOARD SLOTS • N INTERROGATE KEY

256 ▾

rickshaw-0 ▾

Filing cabinet — KV-cache  $O(n)$  • exact

Slots held: 500

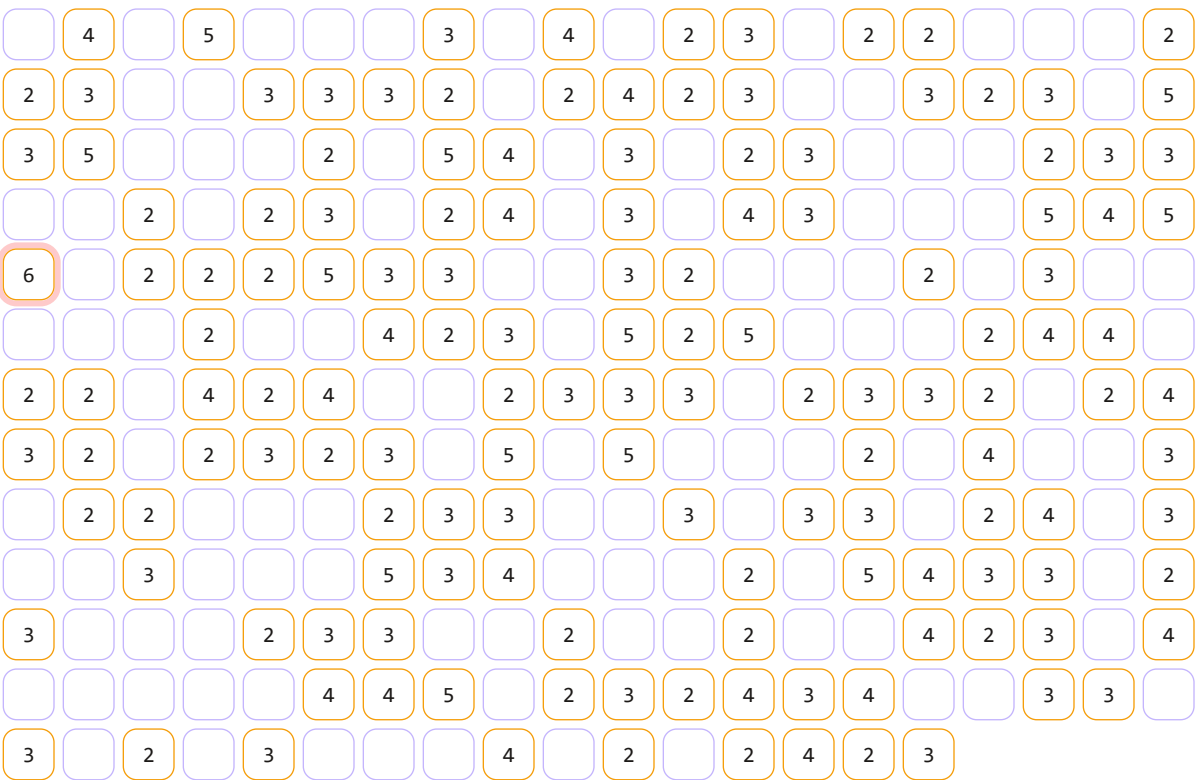


+380 more drawers — it never stops growing

Grows one per token. Retrieval: dot-product over everything. Cost: you pay for every old token, forever.

Whiteboard — recurrent state 0(1) · lossy

Slots: 256 · overwrites: 291



Yellow = someone got erased here. Red = currently lying to you. Click a cell to cross-examine.

Truth beside estimate

| Key                     | Truth | KV says | Whiteboard says | Who stole it?                              |
|-------------------------|-------|---------|-----------------|--------------------------------------------|
| rickshaw-0<br>drawer 80 | 41    | 41      | 92 X erased     | overwritten by cat-24, banyan-21, cloud-18 |

Whiteboard recall: 41.8% (209/500) · Filing cabinet: 100% · Overwrites: 291

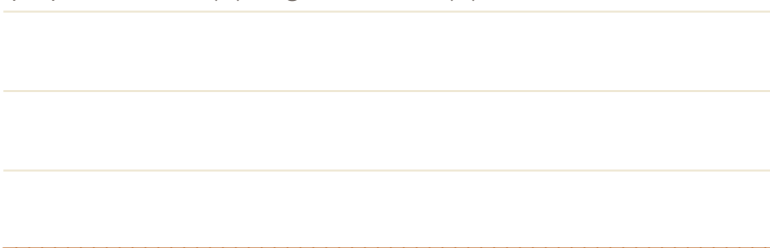
THE BILL · LIVE CURVES

Exactness sends an invoice.

Left: drawers you have to buy. Right: answers you get wrong. Both drawn live from your N — and yeah, they print fine now, we learned that the hard way (canvas, never again).

## Slots vs tokens

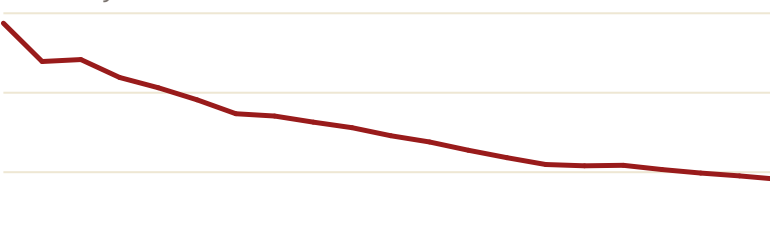
purple climbs  $O(n)$  · green flat  $O(1)$



Right now: cabinet **500** drawers vs whiteboard **256**. At 500: 500 vs 256.

## Whiteboard accuracy vs stream length

accuracy caves as  $n \gg N - \text{live}$



Right now  $n=500$ ,  $N=256 \rightarrow$  **41.8%** correct. Flood it and watch this dive.

BDH MODULE · THE PART WE ACTUALLY SWEATED OVER

## Okay, so the real machine did the whiteboard thing too.

**Who:** Dragon Hatchling (BDH), Pathway's weird-and-cool Post-Transformer. Quick note so nobody gets yelled at in Q&A: **not** a Mamba-style SSM — BDH-GPU is its own GPU-friendly rewrite from ReLU low-rank maps + linear attention. BDH doesn't keep drawers; it scribbles into **synapses while reading**. The one equation we're confident enough to frame (Eq.5):

$\rho_{t-1,l} = \sum_{\tau < t} v_{\tau,l-1}^* x_{\tau,l}^T U^{t-\tau}$  — BDH-GPU attention state, Eq.5: fixed-shape matrix, additive updates,  $O(1)$  in length

Same deal as your whiteboard upstairs, just with learned keys, a decay  $U$ , and ~5% of neurons awake at a time. If you want the full math: Kosowski et al. §3.2 Eqs.4–5, Hebbian lens 6–8, toy code App.E. We read it so you can skim it.

**Receipt #1 · vendor told us, we didn't verify:** 134M BDH gets **95% at 32K → 82% at 128K** on BABILong QA1–QA5 (pathway.com). Holds long, sags under load — exactly your toy's curve, at real scale. Pending contamination / independent / leaderboard review. A benchmark is not a deployment, we'll say it in the viva too.

**Receipt #2 · the cousin BDH-CQ:** 150M, **29.5% pass@2 on ARC-AGI-1 @ ~\$0.0007/task** (arXiv:2608.09888). Thinks in latent space, no chain-of-thought essays — cheap, not champion. Self-reported, weights unreleased.

**Don't leave with the wrong lesson (we almost did):** this forgetting is silent — no "sorry, forgot" flag. And remembering for one chat  $\neq$  learning forever. Getting fast memory into slow weights is still unsolved. Also please, never call BDH an SSM in front of the judges.

#### LANDSCAPE IN 30 SECONDS

### Who pays what?

| Family          | Memory                            | Recall           | Bill                        | Why care                     |
|-----------------|-----------------------------------|------------------|-----------------------------|------------------------------|
| Transformer KV  | $O(n)$                            | <b>perfect</b>   | $O(T^2)$ , OOMs             | gold standard, expensive     |
| Mamba SSM '24   | $O(1)$ gated                      | lossy far        | cheap                       | throughput king, weak needle |
| HOLA '26        | $O(1)$ +cache                     | <b>recovered</b> | cheap+cache                 | hippocampus for surprises    |
| <b>BDH / CQ</b> | <b><math>O(1)</math> synaptic</b> | lossy, sags      | <b>lowest \$/task shown</b> | memory = reasoning fabric    |

## 60-SECOND CHECK • BE HONEST

**Bet you can't call the next move.**

No marks, no tracking. Judges will basically ask you this in the viva, so practice getting it wrong here first.

- ☐ **n=500, N=16:** whiteboard stays perfect because hashing is uniform.
- ☒ **n=500, N=16:** whiteboard collapses; too many keys share too few slots.
- ☐ **KV-cache:** forgets oldest tokens first to stay  $O(1)$ .

**Right. Now prove it: hit “Flood it” and watch recall crater.**

## RECEIPTS • VERIFY, DON'T TRUST

**Sources & caps.**

1. Kosowski et al. *Dragon Hatchling: Missing Link between Transformer and Brain*. arXiv:2509.26507 (30 Sep 2025). Eqs.4–8, BDH-GPU App.E. [arxiv](#)
2. Engdahl, Kosowski et al. *BDH-CQ: In-Context Learning with Recurrent Latent Reasoning*. arXiv:2608.09888 (10 Aug 2026).  $S_t = U(S_{t-1}, D_t)$ . [arxiv](#)
3. Cui. *Hippocampus for Linear Attention*. arXiv:2607.02303 (2 Jul 2026).  $O(1)$  but lossy; overwrites. [arxiv](#)
4. Gu & Dao. *Mamba*. 2024. SSM baseline — related, distinct from BDH.

**Caps, honestly:** toy = string-hash % N, int values, exact-match. No learned keys, decay, sparsity. Small-N curves are live; BABILong/ARC are precomputed vendor numbers. Toy  $\neq$  official BDH. Code: team-written, MIT. AI-assisted (Muse Spark); team defends all of it.

Made at 2am by Cloud 9 (IIT KGP, DataForge) — if something looks off, it's probably our hash function, tell us in Q&A. Single file, no login, no secrets · [repo](#) · [live](#)